

# Literature-implied partial answer: nonseparable Bourgain-Pisier $\mathcal{L}_\infty$ -spaces

Run `fa_banach_001`; source arXiv:1709.09646; supporting arXiv:2604.11832

## Status

### `literature_implied_answer` (partial subcase).

Ferrando–Kakol–Lopez-Pellicer–Sliwa, arXiv:1709.09646, record Mazur’s separable quotient problem in the Introduction:

Does every infinite-dimensional Banach space have an infinite-dimensional separable quotient?

This note records a known-literature answer for the class of nonseparable Bourgain-Pisier  $\mathcal{L}_\infty$ -spaces produced by Lopez-Abad’s construction.

## Supporting Result

Patri, arXiv:2604.11832, proves the following criterion in Theorem `general`: if  $Y$  is an  $\mathcal{L}_\infty$ -space containing a subspace  $X$  such that  $Y/X$  is infinite-dimensional and has the Schur property, then  $Y$  admits a quotient isomorphic to  $c_0$ .

The same paper then applies this criterion in Corollary `lopezabad`: if  $X$  is any infinite-dimensional Banach space and  $Y$  is any nonseparable Bourgain-Pisier  $\mathcal{L}_\infty$ -space over  $X$  constructed by the Lopez-Abad method, then  $Y$  admits  $c_0$  as a quotient. Since  $c_0$  is separable and infinite-dimensional, this gives an affirmative answer to Mazur’s problem on that class.

## Identification

The source survey asks the general separable quotient problem and lists known positive classes such as reflexive spaces, weakly compactly generated spaces, dual spaces, and spaces with suitable copies of  $c_0$  or  $\ell_1$ . Patri’s 2026 theorem supplies another positive class: nonseparable Bourgain-Pisier  $\mathcal{L}_\infty$ -spaces arising from Lopez-Abad’s extension method.

The supporting paper explicitly frames its result as resolving Mazur’s problem for that class, but it is not an exact later answer to a numbered question in arXiv:1709.09646. The connection is therefore recorded as an agent-identified literature implication for a partial subcase, not as an original result and not as a full solution of the Mazur problem.

## Evidence

The source problem appears in arXiv:1709.09646, Introduction, Problem `mazur` in the source TeX lines 161–163. The supporting theorem is arXiv:2604.11832, Introduction, Theorem `general`; the specific Lopez-Abad application is Corollary `lopezabad`. The supporting paper’s Conclusion again states that Mazur’s problem is resolved for nonseparable Bourgain-Pisier  $\mathcal{L}_\infty$ -spaces constructed by the Lopez-Abad method.

The packet includes local copies of both PDFs:

`source_paper.pdf`,      `supporting_paper_2604.11832.pdf`.

## Scope

This packet does not settle the general separable quotient problem. It also does not address the equivalent Schauder-basis quotient problem, the quasi-complement formulation, Sliwa's strongly normal sequence question, or the survey's separate prompt asking for non-reflexive spaces without a copy of  $\ell_1$ . Its scope is exactly the  $\mathcal{L}_\infty$ -with-Schur-quotient criterion and the Lopez-Abad nonseparable Bourgain-Pisier class covered by the supporting paper.

## Search Notes

Local index searches for arXiv:1709.09646 and for the phrases separable quotient, quotient problem, strongly normal, Bourgain-Pisier, Lopez-Abad, and  $\mathcal{L}_\infty$  found no prior packet for this subcase. Web searches on 2026-06-16 found arXiv:2604.11832 and no full resolution of the general Mazur problem.

## References

- [1] J. C. Ferrando, J. Kakol, M. Lopez-Pellicer, and W. Sliwa, *On the separable quotient problem for Banach spaces*, arXiv:1709.09646.
- [2] K. Patri, *Mazur's Separable Quotient Problem for Nonseparable Bourgain-Pisier  $\mathcal{L}_\infty$ -Spaces*, arXiv:2604.11832.
- [3] J. Lopez-Abad, *A Bourgain-Pisier construction for general Banach spaces*, Journal of Functional Analysis 265 (2013), 1423–1441.